

Compact Representations of Human Interception Movements

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1 Introduction

Interception movements are made constantly, yet motor control lacks robust models of their detailed kinematics. Existing models of interception continuously update predictions using feedforward or feedback strategies [1]. However, they lack a compact parametrization for stable individual differences (motor signatures). Because perception and movement are stochastic, the modelling target is a distribution of movement features rather than one deterministic trajectory.

We asked whether a participant's trials can be compressed into a small continuous latent code that regenerates the distribution of that participant's remaining movements. We evaluate five model families on identical participant-held-out folds, report each prediction task and its metric separately, and add a control in which only the supplied fingerprint varies. The contribution is a matched benchmark with an explicit account of what the representation fails to do.

2 Data and task

Participants slid a finger across a tabletop to control a visual cursor on a display, aiming to reach a target square as a circle moved horizontally into it, requiring rapid interception movements under one second. Finger position was recorded in 3D at 240 Hz with a Polhemus Liberty tracker; target trajectories were sampled at 60 Hz. Only condition 2, free eye movements, was analyzed. Of 4,763 available recordings, four missing-arrival timeouts and 27 trials arriving more than 1 s after the target window were excluded, leaving 4,732 trials from 28 participants.

One event definition is critical. Marker 5 records target appearance, after which the target holds still for a randomized 0.20–0.50 s foreperiod before moving. Behavioral zero-time is target motion onset, recovered from the target trajectory rather than the marker; anchoring to the marker would inject the randomized foreperiod into every initiation-time measurement. The window ends at the last tracker sample, serving as an arrival proxy that precedes the recorded arrival event by an average of 26.2 ms.

3 Methods

3.1 Representation and models

Positions are fourth-order zero-phase Butterworth filtered at 10 Hz, translated to the initial window position and resampled to 100 phase points. The final models use lateral x and forward y . Resampling removes physical duration, so encoders see shape only; movement time and initiation time are withheld from every encoder and predicted separately.

The CVAE encodes 100×2 positions and five condition values through two 256-unit ReLU layers to a diagonal Gaussian posterior [2,3]; its decoder maps a latent draw and conditions to trajectory shape and two log-duration outputs (initiation and movement times).

Spline + PCA is the capacity-matched comparator: a cubic B-spline with five fixed interior knots gives 18 coefficients, compressed by PCA fitted on training participants only to the same latent size, with a separate Ridge regression predicting log-durations.

Two ablations isolate design choices: a conditional autoencoder without the stochastic bottleneck, and an unconditional VAE with conditions zeroed. A condition-only Ridge model provides the floor. The sweep covered $n \in \{2, 3, 4, 8\}$; the planned $n=16$ was not evaluated.

3.2 Participant-held-out protocol

Four fixed folds use 17 training, 4 validation and 7 test participants; every participant is tested once. For each participant fold, neural models are trained across seeds 42, 43, and 44 to evaluate different weight initializations and optimization randomness under identical data splits. Each test participant's trials are split into disjoint context and query halves, stratified by start category and side. The fingerprint is the mean code over a participant's context trials: the mean encoder posterior mean for the neural families, the mean PCA score for Spline + PCA. Around this fingerprint, 120 trajectories are generated using a shared training covariance Σ_{train}^1 , which combines within-participant trial dispersion with mean encoder posterior variance; for Spline + PCA the posterior term is absent, since PCA scores are deterministic. These trajectories are conditioned on task parameters drawn from query trials and compared against the query feature distributions. Query shapes and timings never enter the fingerprint. Paired two-sided Wilcoxon tests are conducted using one seed-averaged value per participant with Pratt zero handling and Benjamini-Hochberg correction [4] within each declared family.

4 Task 1: trajectory reconstruction

Reconstruction measures how well an observed trial is reproduced from its individual latent code, using posterior means and participant-balanced MSE (averaged across trials and seeds per subject).

Table 1. Participant-balanced reconstruction error. Lower is better. The condition-only Ridge has no latent dimension; its result is repeated for reference.

Model	n=3 MSE	n=8 MSE
Spline + PCA	0.097255	0.022772
VAE	0.148299	0.033487
Conditional AE	0.169661	0.025968
CVAE	0.176981	0.032481
Condition-only Ridge	1.233795	1.233795

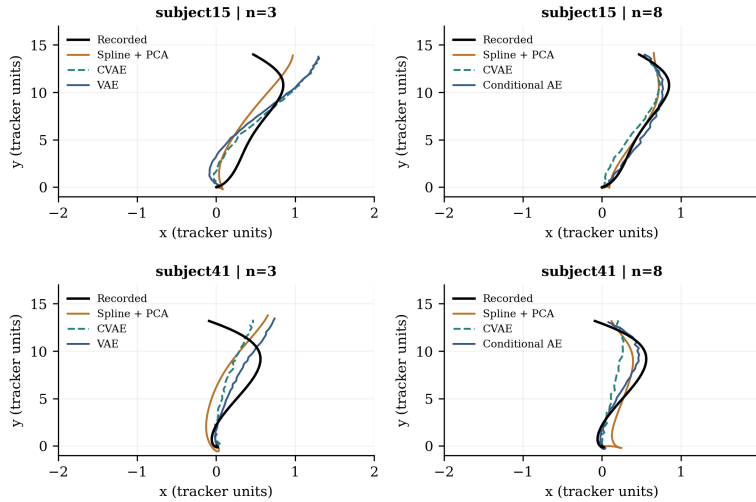


Figure 1. Recorded inputs and decoded outputs at neural seed 42; blue is VAE at n=3 and conditional AE at n=8. All panels share axis limits.

Spline + PCA achieved the lowest error at both capacities. This is expected: PCA is the optimal linear projection under squared error, while the variational models trade reconstruction accuracy for a sampleable latent space. Among neural models VAE was best at n=3 and conditional AE at n=8, demonstrating that conditioning alone does not explain neural performance rankings.

¹ Neural families: $\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\mu_i - \mu_{s(i)}) + \text{diag}(E_{\text{train}}[\sigma_i^2]) + 10^{-6}I$. Spline + PCA: $\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(z_i - z_{s(i)}) + 10^{-6}I$, where z are PCA scores.

5 Task 2: generating participant feature distributions

The generation task evaluates 120 sampled trajectories per participant against their held-out query trials across 11 kinematic features. Evaluations use the mean two-sample Kolmogorov–Smirnov (KS) statistic and two multivariate distance metrics: squared MMD [5] and Energy distance [6]. Crucially, none of these measures represents a point-to-point reconstruction error.

Table 2. Cohort-averaged generated feature discrepancies. Lower is better. All three metrics compare distributions.

Model	n	Mean KS	Energy	MMD squared
Spline + PCA	3	0.325126	0.977433	0.181467
CVAE	3	0.277795	0.745431	0.130935
Conditional AE	3	0.287578	0.801489	0.141410
VAE	3	0.284587	0.747566	0.130298
Spline + PCA	8	0.254794	0.734715	0.115291
CVAE	8	0.219404	0.430451	0.067493
Conditional AE	8	0.258636	0.620200	0.111100
VAE	8	0.215100	0.411000	0.063900

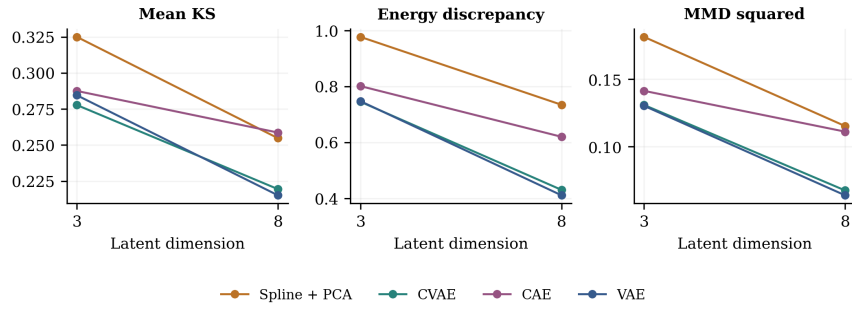


Figure 2. Generated-feature discrepancies at three and eight dimensions. Lines connect tested settings only and do not estimate untested capacities (draft Figure 7).

At $n=8$, the unconditional VAE attains the lowest cohort-average values on all three metrics and serves as the dashboard default for that reason. Crucially, this reflects a numerical preference on the evaluated cohort rather than uniformly significant superiority: of seven CVAE-versus-VAE contrasts at $n=8$, exactly one reaches statistical significance (favoring the unconditional model). At $n=3$, no CVAE improvement over the condition-only Ridge survives FDR correction; at $n=8$, all three metric improvements survive.

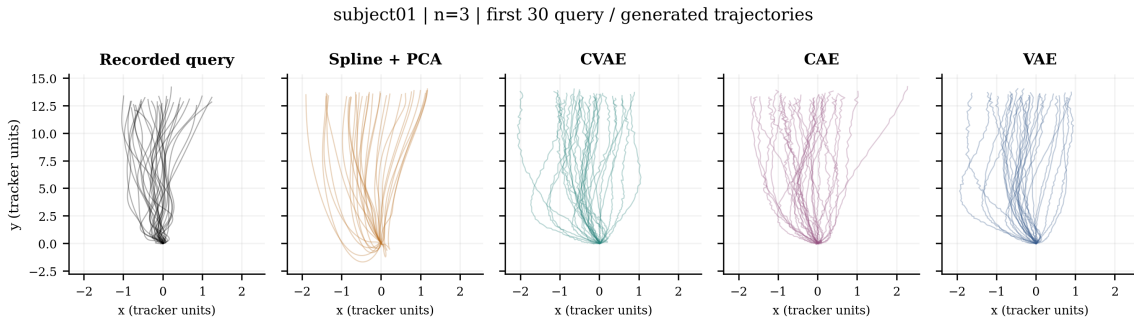


Figure 3. Recorded query and generated trajectories at $n=3$ for subject01; the first 30 available samples in fixed ordering (draft Figure 8).

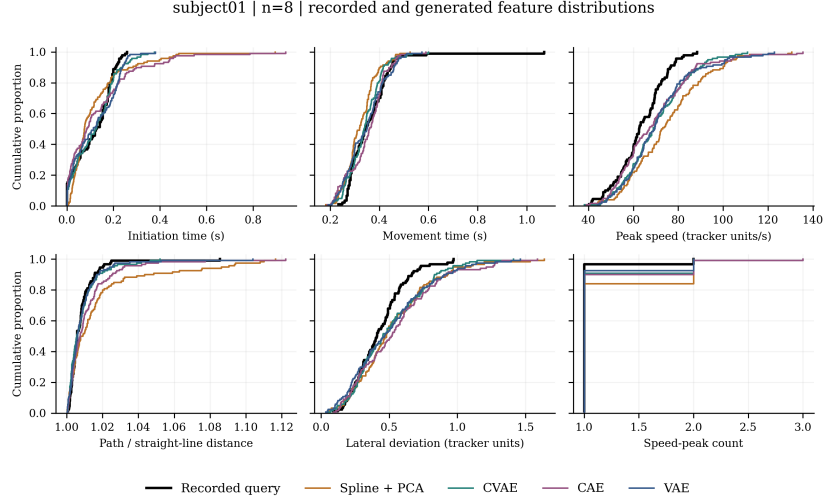


Figure 4. Recorded query and generated feature distributions for the same participant at n=8 (draft Figure 10).

6 Control: contribution of the personal fingerprint

A same-decoder control isolates the effect of the individual fingerprint. Across all 24 CVAE checkpoints ($n \in \{3, 8\}$, 4 folds, 3 seeds), the decoder, 120 condition draws, latent noise, shared covariance Σ_{train} , and distance reference are held fixed; only the supplied center is replaced — either by the overall population mean or by another participant's fingerprint.

Table 3. Fingerprint replacement with the generative procedure held fixed. Lower is better. The other-person row averages donor-specific distances.

Model	Fingerprint center	Mean KS	Energy	MMD squared
CVAE n=3	Own participant	0.278	0.745	0.131
CVAE n=3	Training-population mean	0.311	1.016	0.175
CVAE n=3	Other test participants	0.338	1.273	0.220
CVAE n=8	Own participant	0.219	0.430	0.067
CVAE n=8	Training-population mean	0.290	0.919	0.149
CVAE n=8	Other test participants	0.317	1.196	0.199

The correct fingerprint improves all three distance metrics against both controls across both dimensions (12 contrasts, all $p_{\text{BH}} < 0.0004$). At n=3, the mean KS statistic improves over the population center for 22 of 28 participants and over other-person centers for 27 of 28; at n=8, both counts reach 27 of 28. Because only a single input was varied, this performance gain is directly attributable to the fingerprint. Importantly, this establishes input-sensitivity rather than proving that the fingerprint captures a stable cognitive strategy across sessions.

7 Task 3: recovering physical timing from shape

Physical durations are withheld from every encoder, so predicting them from normalized shape and task conditions constitutes a separate task, evaluated using participant-balanced MAE.

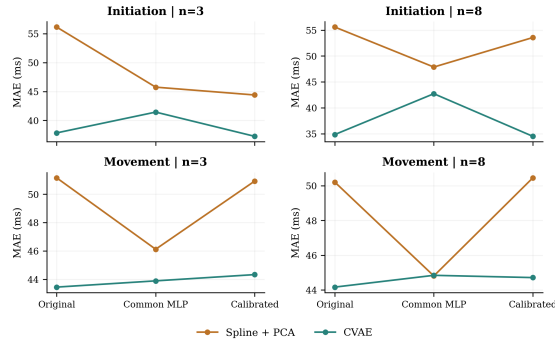
Table 4. Original timing-head errors. Lower is better.

Model	n	Initiation MAE (ms)	Movement MAE (ms)
Spline + PCA	3	56.16	51.16
CVAE	3	37.83	43.45
Conditional AE	3	37.75	44.09
VAE	3	37.92	42.91
Spline + PCA	8	55.58	50.21
CVAE	8	34.84	44.16
Conditional AE	8	39.87	42.67
VAE	8	35.98	42.21

The unadjusted baseline comparison was strictly unmatched: the CVAE timing head used a nonlinear MLP, whereas the Spline + PCA baseline relied on a linear Ridge regression, meaning the contrast evaluated predictor capacity alongside representation quality. Fitting an identical two-layer MLP to frozen latent codes from both representations yields the matched comparison in Table 5.

Table 5. Matched-head timing sensitivities in ms. BH adjustment covers the eight paired participant comparisons in the source draft.

Head	n	Duration	CVAE MAE	Spline MAE	BH-adjusted p
Common MLP	3	initiation	41.43	45.75	0.0076
Common MLP	3	movement	43.89	46.12	0.1872
Common MLP	8	initiation	42.72	47.87	0.0423
Common MLP	8	movement	44.84	44.81	0.8314

**Figure 5.** Timing errors under original, common-MLP and validation-calibrated heads. Points are separate fitted procedures (draft Figure 12).

Under a matched head, the n=8 initiation time advantage decreases from 20.7 ms to 5.1 ms yet remains statistically detectable, whereas the movement-time advantage disappears. Crucially, this remains an imperfect geometry contrast, as CVAE latent codes were trained with joint timing supervision, whereas Spline + PCA codes were not.

8 Task 4: participant identification after enrollment

Each test participant is enrolled using their fingerprint (the mean latent code over context trials); query trials are then assigned to the nearest of the seven enrolled participant centers via Euclidean distance.

Table 6. Identification among seven enrolled test participants, chance 14.3%. Descriptive closed-set results, not identification of unseen identities without enrolment.

Model	n=3 balanced accuracy	n=8 balanced accuracy
Spline + PCA	36.4%	48.6%
CVAE	40.9%	50.7%
Conditional AE	39.0%	46.5%
VAE	40.0%	50.8%

Classification accuracy exceeds chance level ($1/7 \approx 14.3\%$) across all settings. At $n=8$, both variational models reach approximately 51% accuracy, indicating partial separation of participant-specific motor features accompanied by considerable inter-individual overlap.

9 Task 5: predicting behavioural summaries from the code

Validation-tuned Ridge probes predict the means and standard deviations of seven query behaviors directly from context fingerprints. Two baseline controls are evaluated: a population-mean baseline and a direct average of the participant's own context trials.

Table 7. MAE for query-summary targets. Timing in ms; other errors use each feature's units. [EDIT: paste the remaining 10 rows from draft Table 10, or state that the full 14-row table is in the laboratory draft.]

Query-summary target	CVAE n=3	VAE n=8	Direct context	Dev. constant
Initiation time (mean) [ms]	14.778	17.173	9.275	30.381
Initiation time (sd) [ms]	13.717	14.602	10.928	14.028
Movement time (mean) [ms]	36.444	35.394	11.049	48.762
Peak speed (mean) [units/s]	7.744	8.760	1.734	11.876
[EDIT] remaining 10 targets	-	-	-	-

Direct trial measurement outperforms the latent probe on all 14 targets across both latent capacities ($n \in \{3, 8\}$). A nested leave-one-participant-out test evaluated whether the fingerprint adds residual predictive power beyond direct context averaging: across all 28 evaluation cells, none remained significant after correction (mean $\Delta R^2 = -0.017$). Crucially, a positive control verified statistical sensitivity at the same sample size ($N=28$). Thus, given a direct sample of a participant's trials, the latent fingerprint provides no additional information regarding these summary statistics.

10 Feasibility baseline: K-means clustering

An initial feasibility baseline evaluated whether trials are inherently separable by participant. Unsupervised K-means clustering ($K=28$) was fitted on time-normalized trajectories and normalized kinematic features without labels, which were used ex post facto solely to evaluate cluster alignment.

The Adjusted Rand Index (ARI) reached 0.0518 on spatial trajectories and 0.0703 on kinematic features, with corresponding Normalized Mutual Information (NMI) values of 0.1944 and 0.2197. Both metrics significantly exceeded all 200 shuffled-label permutations (empirical $p = 0.00498$), yet absolute ARI values remained near zero. This confirms that participant identity is statistically detectable yet weak—sufficient to motivate a non-linear generative model, but insufficient to form clean spatial clusters. Notably, this descriptive analysis includes all trials, unlike the held-out evaluation framework used above.

11 Control: does explicit task conditioning matter?

Shuffling decoder conditions within each test participant—while holding original latent codes fixed and without retraining—tests whether conditional models actively utilize their conditional inputs.

Table 9. Frozen condition-shuffle diagnostic; the last column counts participants with lower MSE under correct decoder conditions.

Model	n	Correct conditions	Decoder shuffled	Correct better
CVAE	3	0.176981	0.180578	25/28
CVAE	8	0.032481	0.035105	28/28
Conditional AE	3	0.169661	0.171582	26/28

Supplying correct conditions yields superior generation accuracy for 25 of 28 participants at $n=3$ and 28 of 28 at $n=8$, confirming that conditional information is genuinely utilized by the decoder. An architectural audit further revealed that zeroing a conditional network's condition weights recovers the unconditional model exactly. Because the conditional model class subsumes the unconditional solution, incorporating extra inputs does not guarantee empirical superiority under finite training iterations and validation selection.

12 Secondary analysis: minimum-jerk components

Following the proposal, generated movements were benchmarked against Prof. Friedman's submovement pipeline [8] by fitting one to four overlapping minimum-jerk pulses [7]; this is an adapted implementation, not the published scattershot algorithm [9]. Recorded and generated trajectories were originally decomposed with different inputs and optimizer budgets, so part of the measured mismatch was procedural; one identical operator across 4,056 fits reduced the selected-count total variation from 0.353/0.355 to 0.214/0.232 at $n=3/8$, so the earlier values overstated the mismatch.

Table 10. Participant-balanced matched component discrepancies. The reference is the 95th percentile of an empirical resampling distribution, not a formal threshold.

Discrepancy	CVAE $n=3$	CVAE $n=8$	VAE $n=8$	Reference 95th pct.
Fit error KS	0.339	0.308	0.328	0.282
First duration KS	0.455	0.409	0.406	0.283
First amplitude KS	0.377	0.351	0.362	0.283
Count total variation	0.214	0.232	[EDIT]	0.146

All discrepancies remain above the sampling reference and all 4,056 selected fits converged, but no $n=3$ versus $n=8$ contrast survives correction, and the selected count is far more sensitive to the timing crop and filter cutoff than to optimizer budget. The analysis is dominated by fitting assumptions and remains secondary.

13 Limitations

While the study cohort ($N=28$) provided sufficient statistical sensitivity for within-subject controls, relying on a single dataset limits generalizability across broader demographic or clinical populations. Furthermore, cross-validation folds share training participants; consequently, evaluation errors are not strictly independent, rendering p-values somewhat optimistic [10]. Additionally, context and query trials originate from the same recording session, meaning latent fingerprints may partly reflect session-specific factors—such as posture or tracker calibration—rather than temporally stable individual traits.

14 Lessons learned

Audit evaluation code as strictly as model code: Our evaluation metric was initially normalized using evaluation data rather than a fixed training baseline. This caused zero-variance features to divide by near-zero values, creating an adaptive yardstick. Resolving this required recomputing all 124 evaluation checkpoints.

Ensure fair comparisons before drawing conclusions: Apparent model advantages in Sections 7 and 12 actually stemmed from differences in classifier capacity. Standardizing evaluation heads revealed that original metrics reflected procedural asymmetry rather than true representation quality.

Record explicit execution settings over implicit defaults: Key experimental constraints—such as withholding physical timing from both encoders—were managed only in execution scripts rather than class definitions. All effective parameters are now explicitly logged per run.

15 Potential extensions

The most principled extension is an architecture matching the research question. The current model has no person-level variable: the fingerprint is a post-hoc average of trial codes, and nothing in training encourages a participant's trials to cluster. A hierarchical model with an explicit subject latent and a per-trial residual would express that story directly.

Three cheaper extensions follow. The timing weight of 20 was never tuned yet sits on the reconstruction-versus-timing trade-off that decides the comparison. Retraining with timing supervision disabled would isolate representation geometry from the training objective. And the encoder flattens 100x2 into 200 numbers, discarding temporal ordering; a convolutional encoder would exploit the smoothness the spline basis exploits by construction.

For any future recording the decisive experiment is cross-session enrollment: splitting context and query trials across different sessions would test whether the fingerprint is a stable personal trait, the one claim this dataset cannot support.

16 Deliverable: interactive dashboard

The deliverable is a Streamlit dashboard supporting CVAE and spline + PCA at $n=2,3,4,8$ and unconditional VAE and conditional AE at $n=3,8$. It exposes generation from a chosen fingerprint, held-out validation examples, benchmark tables, latent associations and diagnostics, and runs without the private recordings. Live generation uses fold-0 seed-42 references while benchmark tables aggregate repeated runs, as labelled in the interface. A planned second mode accepting new CSV recordings was not implemented.

17 Conclusion

A participant's context-derived fingerprint measurably improves conditional trajectory generation under strict control where all other components remain fixed. However, a capacity-matched Spline + PCA baseline reconstructs trajectories with higher fidelity, an unconditional model ablation matches or exceeds conditional generation accuracy, and direct averaging of a participant's context trials predicts behavioral summaries more effectively. Ultimately, the learned latent representation functions as a useful conditioning signal for generative modeling, rather than a validated low-dimensional account of individual interception strategy.

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